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LAB 8: VELOCITY PROFILE

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ABSTRACT

This experiment investigated the velocity distribution downstream of a cylindrical rod array within the TecQuipment TE93 Cross-Flow Heat Exchanger. A pitot probe was positioned vertically within the working section to measure total and static pressures at multiple locations, which were then converted into local air velocities using Bernoulli's principle. The goal was to determine how the presence of cylindrical rods alters the velocity profile and to visualize the resulting flow structure. The TE93 apparatus provides static and total pressure readings, a movable pitot assembly, and controlled airflow through a motor-driven fan. Results showed a distinct velocity pattern characterized by peak velocities between rod gaps and significant reductions or even flow reversal behind the rods due to wake recirculation. Overall, the experiment demonstrated key fluid dynamic behaviors in obstruction based cross flow, validated theoretical expectations of wake development, and turbulent mixing.

Keywords: Velocity profile, pitot tube, static pressure, total pressure, cross-flow, ideal gas, incompressible, Bernoulli, turbulence, wake region, pressure differential.

Nomenclature

P_t	*Total (pitot) pressure (Pa)*
P_d	*Downstream Static Pressure (Pa)*
P_A	*Barometric (atmospheric) pressure (Pa)*
ρ	*Air density (kg/m3)*
V_2	*Downstream velocity (m/s)*
R	*Gas constant for air (J/kg*K)*
T_1	*Ambient air temperature (K)*
Δp	*Pressure difference (pt-Pa*pd)*

1. INTRODUCTION

Velocity distributions are an essential part of understanding fluid flow in cross-flow heat exchangers, rod bundles, and similar obstruction-based geometries [1][2]. When air flows across a bank of cylindrical rods, wakes develop behind each rod, creating velocity deficits and asymmetry in overall flow profile. The TecQuipment TE93 Cross-Flow Heat Exchanger is equipped with a precision pitot assembly that traverses vertically within the working section, enabling measurements at controlled positions downstream of the rod array [3].

By converting pressure differences recorded at each vertical probe position into local velocity, the velocity profile can be constructed. This profile provides insight into wake development, flow separation, and mixing patterns caused by rod interference. Understanding these effects is important for predicting pressure losses, turbulence behavior, and overall flow distribution in cross-flow systems. The objective of this experiment was to map the velocity field downstream of the rods and interpret how the rod geometry influences flow characteristics

2. THEORY

The TE93 uses a pitot probe to measure total pressure P_T, while static pressure P_d is obtained from a downstream wall pressure tapping. The velocity at each measurement point is calculated from the Bernoulli equation for incompressible flow [4], expressed in terms of measured pressure difference and air density. Air density is calculated assuming ideal-gas relation based on measured barometric pressure and ambient temperature.

Governing equations:

$$V_2 = \sqrt{\frac{2(p_t - p_d)}{\rho}} \tag{1}$$

where
$$\rho = \frac{p_A}{RT_1} \tag{2}$$

Here V_2 is the local downstream velocity at the probe location, ρ is the air density, p_T is total pressure, and p_d is static pressure. The pitot tube senses the velocity head, which is proportional to V_2^2. Rearranging the Bernoulli equation yields the expression in Eq. (1).

In regions directly behind the rods, flow separation and recirculation can reduce local total pressure or even produce negative differential pressures relative to the reference tapping [3]. In those zones, the computed velocity can be negative, indicating reverse flow within the wake.

3. METHODS

3.1 Preparation

The TE93 Duct Assembly and Control and Instrumentation Unit were positioned on a level workbench with the duct assembly aligned to allow clear access to the working section and pitot assembly. All aluminum rods were fitted into the working section to form a full bank of rods, matching the standard configuration for Experiment 3 in the user guide [3].

The pitot assembly was mounted in the downstream position, with the probe oriented to face upstream toward the rod bank depicted in *Figure 1*. The downstream static pressure tapping at the base of the working section was connected to the (-) port of Differential Pressure Input 1 on the Control and Instrumentation Unit. The pitot line was connected to the corresponding (+) port. The digital position indicator on the pitot assembly was switched on, and the probe was carefully lowered until it hovered slightly off the bottom of the working section. This location was taken as the datum, and the indicator was zeroed.

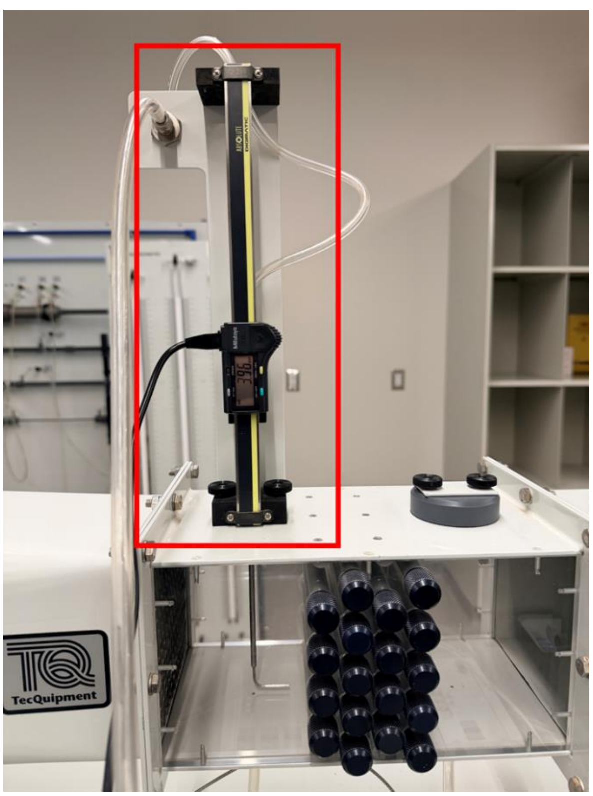

Figure 1 Pitot Assembly Mounted in Downstream Position

3.2 Data Collection Procedure

With the pitot datum set, the airflow valve on the exhaust duct was opened fully (100% position) and the fan was switched on using the Control and Instrumentation Unit. Ambient air temperature T_1 and barometric pressure p_A annotated in the instrument's user guide [3] were utilized as our conditions for *Table 1* and further our sample calculations in 4.2.

The pitot probe was raised from an indicated height of 4 mm above the datum to 118 mm in increments of 2 mm, as specified in the TE93 manual [3]. The actual probe position in the working section is 1 mm greater than the indicated height due to the pitot tip geometry, so the actual measurement positions ranged from 5 mm to 119 mm measured from the floor of the working section shown in *Table 1*. At each height, the differential pressure reading ($P_t – P_d$) was recorded from Differential Pressure Input 1 on the Control and Instrumentation Unit.

At each probe height, the differential pressure reading ($p_t - pd$) was recorded from Differential Pressure Input 1. If negative pressure differences occurred, the magnitude was recorded and later treated according to the velocity sign convention described in Section 3.3.

3.3 Data Reduction and Calculations

Data reduction began with the calculation of air density ρ using the recorded barometric pressure and ambient temperature. The ideal gas relation was applied using equation (2). For each probe location, the local velocity V_2 was then determined from the measured pressure difference $\Delta p = p_t - p_d$. Taking air to be incompressible within the working section of the instrument [3], the Bernoulli-based velocity equation was applied to find V_2 from (1). Of note, for locations where Δp was negative, the computed velocity was additionally observed to take on a negative value to represent reverse flow. *Figure 2* displays the resulting velocity values that were paired with their corresponding actual probe heights, and a velocity profile plot was prepared with the velocities on the y-axis against the probe position on the x-axis respectively.

4. EXPERIMENTS

4.1 Results

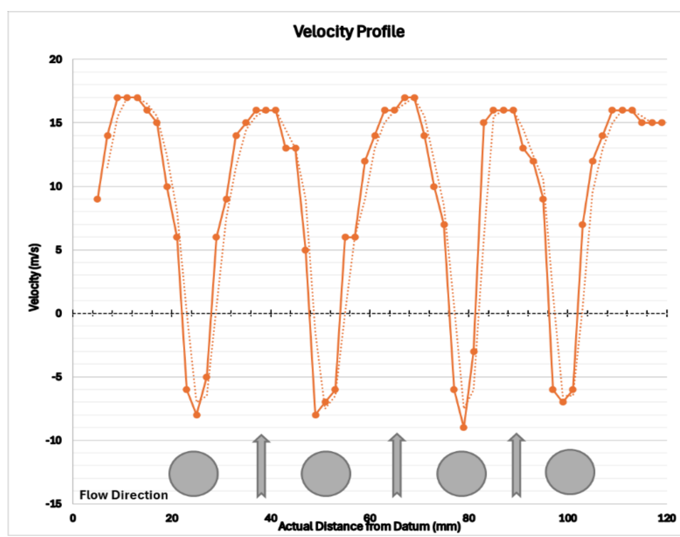

Figure 2 Velocity Profile of the Cross-flow Heat Exchanger

The velocity profile measured downstream of the cylindrical rod bank in *figure 2* showed clear non-uniform distribution resulting from wake formation behind the rods. The highest positive velocities occurred approximately at the midpoints between rows of rods where the airflow was accelerated through the open gaps. In contrast, regions directly behind the rods experienced significantly reduced velocities, with some locations indicating small negative velocities associated with recirculating wake flow.

The velocity profile was generally symmetric about the vertical centerline when plotted, consistent with the symmetric rod arrangement and uniform inlet conditions. Near the lower and upper walls of the working section, the velocity decreased slightly due to boundary layer effects and increased flow resistance. Additionally, the transition from high-velocity regions between the rods to low-velocity wake zones occurred gradually rather than abruptly, indicating a mix of laminar and turbulent flow behavior downstream of the obstructions. The presence of localized negative velocities near the wake region further supports the occurrence of flow separation and recirculation behind the rods. Together, these results confirm the expected characteristics of obstruction-induced cross-flow and demonstrate how rod geometry influences the resulting velocity field.

Table 1 Velocity Profile Downstream of Rod Bank

Indicated Position (mm)	Actual Position (mm)	Δp (Pa)	Velocity (m/s)
4	5	45	9
6	7	108	14
8	9	164	17
10	11	164	17
12	13	166	17
14	15	156	16
16	17	125	15
18	19	64	10
20	21	23	6
22	23	-24	-6
24	25	-37	-8
26	27	-12	-5
28	29	18	6
30	31	47	9
32	33	117	14
34	35	127	15
36	37	147	16
38	39	157	16
40	41	154	16
42	43	106	13
44	45	93	13
46	47	14	5
48	49	-41	-8
50	51	-26	-7

4.2 Sample Calculations

A sample calculation is shown below for a representative measurement point. Assuming the following conditions [3] at one probe location:

$p_A = 101325\ Pa$
$T_1 = 295\ K$
$\Delta p = p_t - p_d = 20\ Pa$
$R = 287\ J/kg \cdot K$

First, compute the air density ρ:

$$\rho = \frac{p_A}{(R\,T_1)} = \frac{101325}{(287 \times 295)} \approx 1.20\ kg/m^3$$

Then, compute the local velocity V_2 from (1):

$$V_2 = \sqrt{\left(\frac{2\,\Delta p}{p}\right)} = \sqrt{\left(\frac{2 * 20}{1.20}\right.} \approx 5.77\ m/s$$

This calculation demonstrates how the measured pressure differential is converted into local air velocity. The same process was applied to all probe positions to obtain the complete velocity profile downstream of the rod array. For positions where the differential pressure (Δp) was negative, the magnitude of Δp was used in the calculation, and a negative sign was applied to the resulting velocity to represent reverse flow within the wake region.

4.3 Discussion

The measured velocity profile aligns with the expected flow patterns for cross-flow over a bank of cylindrical rods. As the air moved through the gaps between rods, it accelerated, producing distinct velocity peaks in those regions. Conversely, the flow separated behind each rod, forming wake zones characterized by reduced velocity and occasional reverse flow. These observations are consistent with bluff-body flow behavior [4] where separation and recirculation occur downstream of obstructions. The shape of the measured profile also reflects the presence of turbulence and mixing generated by the rods. Instead of displaying a sharp transition between high- and low-velocity regions, the flow gradually shifted from accelerated zones into wake regions. This smoothing effect suggests a combination of laminar and turbulent behavior downstream of the rod bank, consistent with expected conditions inside the TE93 working section [3].

Although the overall velocity trend matched what was expected, there were some noticeable differences between the measured data and the ideal flow pattern. One of the main reasons for this is the alignment of the pitot probe. Because the pitot tube must face directly into the airflow to measure total pressure correctly, even a small angle or shift while adjusting the height could affect the reading, especially in areas where the flow direction changes inside the wake. Some fluctuation in the differential pressure readings was also observed during data collection. This may have been caused by small vibrations in the probe, disruption of the pitot's collinearity with inlet air flow during height adjustments, and natural unsteadiness within the wake flow behind the rods.

Another source of variation may have come from changes in temperature throughout the experiment. Since the air density was calculated from ambient conditions, any shift in temperature or barometric pressure would slightly affect the final velocity values. The pressure sensor also takes a moment to stabilize, and recording measurements too quickly may have added extra variability.

These issues could be reduced by taking multiple readings at each height and averaging the results, allowing the pressure reading more time to stabilize before recording, and ensuring the pitot probe is aligned carefully before each measurement. Using a fixed mechanical adjustment or alignment guide rather than hand-positioning the probe would also help improve consistency in future measurements.

5. Conclusion

The objective of this experiment was to measure the velocity profile downstream of a cylindrical rod bank using the TecQuipment TE93 Cross-Flow Heat Exchanger and evaluate how the presence of obstructions affects airflow behavior. The measured profile showed a clear pattern of accelerated flow in the open gaps between rods and reduced or reversed flow within wake regions behind the rods which aligned well with expected cross-flow characteristics. While minor differences existed between the measured and ideal profiles likely caused by pitot-probe malalignment, sensor stabilization time, and environmental variations, the overall results still reflected the predicted flow structure. These findings confirm that the pitot-static method was effective for capturing velocity changes and identifying wake formation in obstruction-based flows.

Beyond verifying theoretical behavior, this experiment demonstrates how flow irregularities, turbulence, and wake patterns play an important role in real engineering systems. Cross-flow velocities and wake behavior directly influence heat transfer efficiency and pressure losses in applications such as heat exchangers, nuclear reactor rod bundles, air-cooled condensers, and turbine blade cooling channels. Understanding how geometry affects velocity distribution helps engineers design systems that balance performance, efficiency, and reliability. Overall, this experiment successfully met its objectives and provided meaningful insight into how obstructions modify airflow and how these effects can be measured and analyzed in practical engineering scenarios.

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In Depth Velocity Profile Chart -

Indicated Position (mm)	Actual Position (mm)	Δp (Pa)	Velocity (m/s)
4	5	45	9
6	7	108	14
8	9	164	17
10	11	164	17
12	13	166	17
14	15	156	16
16	17	125	15
18	19	64	10
20	21	23	6
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44	45	93	13
46	47	14	5
48	49	-41	-8
50	51	-26	-7
52	53	-19	-6
54	55	19	6
56	57	20	6
58	59	88	12
60	61	123	14
62	63	149	16
64	65	165	16
66	67	161	17
68	69	118	17
70	71	26	14
72	73	-23	10
74	75	-43	7
76	77	-6	-6
78	79	44	-9
80	81	110	-3
82	83	139	15
84	85	147	16
86	87	157	16
88	89	145	16
90	91	94	13
92	93	83	12
94	95	52	9
96	97	-23	-6
98	99	-32	-7
100	101	-18	-6
102	103	26	7
104	105	83	12
106	107	118	14
108	109	146	16
110	111	151	16
112	113	142	16
114	115	136	15
116	117	139	15
118	119	128	15